

electronics products in the previous year.

Tools and solutions available for design engineers

The continuously-evolving sensor world has become so important that lots of initiatives are being taken in this field. Every now and then we see many conferences, expos and seminars to promote the knowledge of sensor-based technologies and products. The worldwide market for sensors also includes kits, modules and training programmes to help the young engineers take initiatives and innovate new products.

There are quite a few development boards available in the market for a geek who wants to try out a few sensors and build some interesting applications. Harsha Bagur V., group manager, In-vehicle Infotainment, Robert Bosch, shares some valuable information on this. "Popular development boards like Arduino, Raspberry Pi and BeagleBone are being increasingly used by DIY enthusiasts. Arduino, however, is suited for simpler and less process-intensive and interactive applications. Both Raspberry and BeagleBone come with popular OS distributions like Linux and Android. BeagleBone, especially, does have multiple options for external connectivity and hence there are possibilities to realise use cases with multiple external sensor connectivity."

Priya Kuber, managing director at Arduino India, reminds us that all programming boards these days have advanced software libraries to detect sensors. A sensor usually sends a string of data, and how to interpret the data into a digital form and make use of those values to get desired result with a programming board is the job of a design engineer these days. "Since everything is based on data, I would say that the new 'tools and initiatives' are the software libraries that run on the board. For example, Capacitive Sensing Library is a library that converts ordinary aluminium foil into a useful calibrated sensor," she adds.

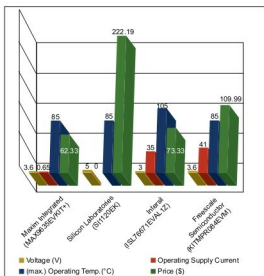


Fig. 3: Comparison of four optical sensor development tool kits

Availability of standalone sensor evaluation boards as commercial off-the-shelf units have immensely helped design engineers integrate the same in their designs at prototyping/proof-of-concept stage and design products with use cases that were never thought of earlier. "One of the biggest gainers of such sensors is wearable computing category—smart watch, wrist bands for health and wellness, sports performance monitoring, etc. And, sensor providers publish APIs/trainings that enable designers adapt to a particular sensor quickly," says Dhaval Vasa, project manager at eInfoChips.

Some sensor-based kits and modules available with their price list are listed in Table II.

Apart from these modules, there are some development tools of popular brands, based on some particular types of sensors, that can act as a learning source for the young engineers and also help them in various innovations.

How to select a sensor for your application

A sensor should be selected depending on the application in mind. Sensors are of two types: limit detector and qualitative and quantitative analysis measuring element. "Limit detection type of sensor gives out logic 1 or 0. This operates as a watchdog in the system and annunciates in fast response at occurrence of moni-

tored parameter. Specifically, this could be an active element used for a particular purpose as a feedback circuit in a power supply, or a Hall effect switch detecting the presence of magnetic field of rated magnetic flux and direction. The other example is a PIR detector detecting presence of an object and movement of the same. Qualitative and quantitative analysis type of sensor could do the job rather slowly and deliver its response in a required electrical form," explains P. Chow Reddy, R&D manager, Power Division, ICOMM Tele Ltd.

There are certain things which a designer has to ensure so that his product does not fail due to a faulty selection of the sensor involved.

Identifying the sensing parameter. The parameter could be in any form such as voltage, current, frequency, temperature, pressure, light, touch, presence, sound and chemical reaction to name a few. Any of these needs to be converted to electrical form for electronic analysis. "A thermistor varies its resistance depending on the thermal stress over it. Thermistor as an element in a potential divider circuit results out a temperature corresponding to the potential across it. An engineer working on this needs to analyse the range of operation of thermistor, way to get safe operating range of the sensor, observe the performance of the thermistor to see what time it takes to convert the parameter, and the linearity and response of the sensor for the physical changes occurring across it," explains Chow Reddy.

Dhaval Vasa gives us advice on the considerations to be made whilst selecting an IR sensor for sensing the proximity of objects. "If you take an example of a proximity sensor (IR-based), which emits an electromagnetic field or a beam of electromagnetic radiation, and looks for changes in the field or return signal, one essentially needs to carefully consider the 'sensing frequency' at which sensing data is provided, 'range' that gets covered in form of angle/distance